

Student Project Showcase Portal

Scenario

The Faculty of Computing wants a web application where students can showcase their projects and recruiters can browse them.

The system should allow:

Users

- Login using Google OAuth.
- Create, edit, and delete project posts.
- Upload a project thumbnail/image.
- View projects created by other students.

Recruiters

- Browse projects.
- "Like" projects.
- Follow students.

Admin

- View all users and projects.
 - Remove inappropriate projects.
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Functional Requirements

1. Authentication (OAuth)

Users must sign in using Google OAuth.

Required information:

- Name
- Email
- Profile picture

Store user details in the database.

2. Project Management (Request/Response)

Students should be able to:

- Create project
- Update project
- Delete project
- View project list

Example APIs:

POST /projects

GET /projects

GET /projects/:id

PUT /projects/:id

DELETE /projects/:id

3. Event-Driven Component

Whenever:

- A project is created
- A project receives a like

Generate an event.

Example:

ProjectCreated

ProjectLiked

The event should trigger:

- Notification creation
- OR
- Activity log entry

Example:

John liked your project.

Students can implement this using:

EventEmitter

or a simple pub/sub pattern.

4. Notifications

Display notifications to users.

Examples:

Mary liked your project.

Alex started following you.

Notifications must be generated only through events.

5. Database

Suggested tables:

users

projects

likes

followers

notifications

Technical Requirements

Frontend

Use one:

- React
- Vue
- Angular or any other solution

Must demonstrate:

- API calls
 - Authentication flow
 - Form validation
 - State management
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Backend

Use one:

- Node.js + Express
- NestJS
- Java or any other solution

Must demonstrate:

- REST APIs
 - Middleware
 - Authorization
 - Event handling
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Deliverables

1. Source code
 2. API Documentation
 3. Deployment URL
 4. demo
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Marking Scheme (100 Marks)

Area		Marks
Requirement gathering	10	
Frontend UI	10	
REST API Design	10	
OAuth Implementation	10	

Event-Driven Architecture	15
Database Design	10
Security & Authorization	10
Documentation	5
Additional features	10
Deployment	10